A Comparative Study of Investment Strategies (2016–2025)



QUANTITATIVE FINANCE RESEARCH REPORT

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Executive summary

This project was motivated by a practical investment-management question: does mathematical portfolio optimization reliably improve real-world, out-of-sample performance over a simple diversified allocation once estimation error, risk and trading costs are considered? To investigate, the study constructed four long-only strategies—Equal Weight, Minimum Variance, Maximum Sharpe and Risk Parity—using eight liquid ETFs spanning domestic and international equities, fixed income, real estate and gold.

The main study used a 252-trading-day rolling estimation window and monthly portfolio rebalancing. Performance was evaluated over approximately February 2016 through December 2025 (2,495 trading days), followed by drawdown analysis, historical tail-risk measures, 10,000-path Monte Carlo simulations, hypothetical economic shocks, turnover measurement and transaction-cost sensitivity.

The most important finding was that the simple Equal Weight strategy led the evaluated portfolios on realized annualized return (7.89%) and Sharpe ratio (0.63). Minimum Variance delivered the lowest volatility (1.56%) and shallowest maximum drawdown (−6.00%). The Maximum Sharpe optimizer had the highest annualized one-way turnover (326.54%) and the deepest maximum drawdown (−34.09%); applying a 10-basis-point trading-cost assumption reduced its annualized return from 4.13% gross to 3.48% net. These results illustrate the distinction between optimizing a portfolio on estimated inputs and building a robust investment process.

Evidence qualification: the numerical tables below use the completed-analysis results recorded during the project workflow. An available notebook copy contains a related but not identical implementation, including a different Monte Carlo sampling method. Accordingly, these reported figures should be reconciled against the final exported CSV files and final notebook before public release; they are not presented as freshly recomputed from that notebook copy.

1. Why this project was undertaken

1.1 The investment problem

Investors face competing objectives: seek return, control volatility, withstand market shocks, preserve liquidity and avoid excessive trading. Traditional mean–variance optimization offers a formal solution, but its weights depend on estimated returns and covariances. Small input changes can produce large allocation changes, particularly for

expected-return-sensitive methods. An apparently optimal backtest can therefore disappoint in live implementation.

1.2 Research purpose

The purpose was not to find a universally superior ETF mix. It was to test how different portfolio-construction philosophies behave under a consistent investment universe and out-of-sample process, and to connect return statistics with practical risk and implementation constraints.

1.3 Research questions

- Which strategy achieved the strongest realized return and risk-adjusted performance?
- How did the strategies differ in volatility, drawdown and downside tail exposure?
- Did simulated outcomes and hypothetical stress scenarios expose vulnerabilities not visible in average returns?
- How much did rebalancing activity and assumed transaction costs change the relative attractiveness of strategies?
- What design choices and data limitations restrict the interpretation of the results?

2. Investment universe and data preparation

2.1 ETF selection

ETF	Exposure represented	Reason for inclusion
SPY	U.S. large-cap equities	Core U.S. growth exposure
VXUS	International equities ex-U.S.	Geographic diversification
VWO	Emerging-market equities	Higher-growth and geopolitical risk exposure
IEF	Intermediate U.S. Treasuries	Interest-rate and defensive exposure
LQD	Investment-grade corporate bonds	Credit-spread and duration exposure
VNQ	U.S. real estate investment trusts	Real-estate and rate-sensitive exposure
GLD	Gold	Alternative asset / potential crisis diversifier
SHY	Short-term U.S. Treasuries	Low-duration defensive allocation

The ETFs were selected to represent different economic risk drivers rather than eight variations of the same equity exposure. This breadth makes correlations, concentration

and portfolio construction meaningful. The universe is nevertheless a deliberate research sample, not an exhaustive investable universe.

2.2 Observation period and return construction

The analysis sourced daily historical adjusted-price series covering 2015–2025, providing a one-year initial estimation window before the out-of-sample period. Daily simple returns were calculated as $r(t) = P(t)/P(t-1) - 1$. Alignment of trading dates and missing-data checks are necessary because portfolio weights must apply to returns observed on a common date. Adjusted prices are intended to reflect distributions and corporate actions, subject to vendor conventions.

2.3 Why preprocessing matters

A portfolio can appear unusually strong if returns are misaligned, missing values are inadvertently forward-filled, prices are confused with returns, or future data leak into an optimization window. The research therefore separates data preparation from strategy evaluation and uses prior-window observations to estimate each rebalance decision. Reproducibility also requires fixing ticker order, return convention, date range and annualization assumptions.

3. Exploratory analysis and investment hypotheses

Descriptive analysis examines annualized arithmetic returns, annualized volatility and pairwise return correlations. Annualized volatility is daily standard deviation multiplied by $\sqrt{252}$; the annualized arithmetic mean is daily average return multiplied by 252. These summaries help distinguish assets that offer expected return potential from those that primarily stabilize a portfolio.

Correlations are central because diversification depends on how assets move together, not simply on the number of holdings. Treasury exposures may respond differently from equities during certain risk-off episodes; however, correlations are time-varying and can change sharply during inflation or liquidity shocks. Gold and real estate should not be assumed to hedge every crisis. The exploratory stage therefore generates hypotheses to test, not proof of persistent protective relationships.

4. Portfolio construction: four competing philosophies

4.1 Equal Weight

Equal Weight assigns 12.5% to each of the eight ETFs at each scheduled rebalance. It requires no expected-return or covariance optimization and serves as a transparent, low-complexity benchmark. It is not risk-balanced: high-volatility assets can dominate portfolio risk even when capital weights are equal.

4.2 Minimum Variance

Minimum Variance chooses weights to minimize portfolio variance, $w^T\Sigma w$, subject to weights summing to one and individual weights lying between zero and one. This

strategy focuses on reducing total return variability, often favoring assets with low measured volatility. Its weakness is that it may sacrifice long-run growth and concentrate in low-volatility or duration-sensitive holdings.

4.3 Maximum Sharpe

Maximum Sharpe seeks to maximize $(w^T\mu - r_f)/\sqrt{(w^T\Sigma w)}$, under the same long-only fully invested constraints. Because expected returns μ are noisy, the optimized weights can change dramatically across rolling windows. This strategy provides a direct test of whether theoretically efficient allocations remain attractive out of sample.

4.4 Risk Parity

Risk Parity aims to distribute risk contributions more evenly across assets. A typical asset contribution to portfolio volatility is $RC_i = w_i(\Sigma w)_i / \sigma_p$. Equal risk contribution is different from equal capital weight. It can moderate the dominance of volatile assets but remains sensitive to estimated covariance and the chosen constraints.

4.5 Optimization safeguards

The project used long-only, fully invested portfolios with weights constrained between 0% and 100%. These rules avoid leveraged or short-selling solutions and improve comparability. They do not eliminate concentration, estimation error or regime sensitivity. Optimization convergence, numerical tolerance and fallback handling should be documented in the reproducible notebook.

5. Out-of-sample backtesting design

5.1 Why rolling estimation was used

An in-sample optimum can benefit from information unavailable when a real decision would have been made. To reduce this look-ahead problem, the study used the preceding 252 trading days to estimate weights and then evaluated subsequent returns. Re-estimation occurred monthly, reflecting a practical balance between adapting to new information and limiting turnover.

5.2 Holding-period mechanics

At each rebalance, target weights are established from historical data. Between rebalances, holdings drift with relative asset performance; they are not silently reset to target weights every day. This distinction matters because daily rebalancing can overstate trading activity and alter the strategy's economic meaning. The main results cover approximately 2,495 trading days from February 2016 through December 2025.

5.3 Evaluation conventions

Portfolio performance was summarized using annualized return, annualized volatility, Sharpe ratio and maximum drawdown. Sharpe ratios are sensitive to the assumed risk-free rate and whether return annualization uses an arithmetic or compounded convention.

The performance tables should be interpreted under the conventions implemented in the final analysis rather than as directly interchangeable with every external benchmark.

6. Performance results and interpretation

Strategy	Gross return	Net return (10 bps)	Volatility	Gross Sharpe	Max drawdown
Equal Weight	7.89%	7.87%	9.37%	0.63	−20.74%
Minimum Variance	1.87%	1.86%	1.56%	−0.08	−6.00%
Maximum Sharpe	4.13%	3.48%	10.90%	0.20	−34.09%
Risk Parity	4.97%	4.89%	5.18%	0.57	−14.87%

6.1 Equal Weight: strongest realized reward

Equal Weight had the highest realized gross and 10-bps net annualized returns and the highest gross Sharpe ratio among the four strategies. Its performance is consistent with the possibility that diversification and stable allocations can outperform more complex estimation-driven approaches. The results do not establish that equal weighting will always win or that estimation error alone caused the performance gap.

6.2 Minimum Variance: capital preservation trade-off

Minimum Variance achieved the lowest volatility and smallest observed drawdown, consistent with its explicit risk-minimization objective. However, its 1.87% gross annualized return and negative reported Sharpe ratio reveal a material opportunity cost. This strategy could suit a narrowly defined low-volatility mandate, but its suitability depends on return targets, inflation, funding needs and rate sensitivity.

6.3 Maximum Sharpe: optimization versus robustness

Maximum Sharpe recorded 10.90% volatility, −34.09% maximum drawdown and a 0.20 gross Sharpe ratio. These outcomes were weaker than the optimizer's theoretical goal would suggest. A plausible explanation is instability in estimated expected returns, potentially combined with allocation concentration and market-regime changes; confirming the mechanism would require inspecting the monthly weights and parameter estimates.

6.4 Risk Parity: intermediate risk profile

Risk Parity generated 4.97% gross annualized return with 5.18% volatility and a −14.87% maximum drawdown. It offered a lower-volatility profile than Equal Weight and Maximum Sharpe, with a positive reported Sharpe ratio of 0.57. This supports its role

as a risk-allocation alternative, though it did not lead on absolute return in the observed period.

7. Drawdowns and historical tail risk

Maximum drawdown measures the largest peak-to-trough decline in compounded portfolio wealth. Unlike volatility, it describes a cumulative loss experience that investors can recognize operationally. The study observed drawdowns of −20.74% for Equal Weight, −6.00% for Minimum Variance, −34.09% for Maximum Sharpe and −14.87% for Risk Parity. The large Maximum Sharpe drawdown is particularly important because it indicates that optimizing a historical return-to-volatility ratio does not guarantee a smooth wealth path.

Historical Value at Risk (VaR) at 95% is a loss threshold exceeded by roughly 5% of daily observations under the empirical distribution. Conditional VaR (CVaR), or expected shortfall, summarizes the average loss in that worst tail. Both complement standard deviation but depend on the sampling window and historical return distribution. Exact VaR/CVaR figures should be inserted from the final `portfolio_risk_metrics.csv` rather than reconstructed or guessed.

8. Monte Carlo simulation: range of potential outcomes

8.1 Purpose

Historical backtests show one realized market path. Monte Carlo analysis explores many hypothetical paths under an assumed return-generating process, helping assess dispersion and the chance of losses over a future-like horizon. It does not predict future returns or eliminate model risk.

8.2 Model and implementation

The completed analysis recorded 10,000 one-year simulation paths of 252 trading days, using historical moments under a multivariate normal daily-return assumption. This model preserves an estimated covariance structure but may underrepresent heavy tails, regime shifts and serial dependence. An available notebook copy instead uses an empirical bootstrap for a selected strategy; the final published report and code should consistently identify which method produced each output.

Strategy	Simulated probability of one-year loss
Equal Weight	21.60%
Minimum Variance	11.62%
Maximum Sharpe	37.83%
Risk Parity	17.60%

Maximum Sharpe had the largest modeled loss probability (37.83%), while Minimum Variance had the smallest (11.62%). These probabilities are conditional on the simulation assumptions and estimated historical inputs, not calibrated forward-looking forecasts. The exercise demonstrates how a portfolio that appears efficient on point estimates can still produce a wide distribution of investment outcomes.

9. Hypothetical economic stress testing

Stress testing addresses a different question from Monte Carlo simulation: how would current portfolio weights react to a deliberately severe, specified combination of asset shocks? Three illustrative scenarios were constructed: Equity Crash, Interest Rate Shock and Inflation Shock. They are assumed one-period shock vectors, not reconstructions of particular historical episodes or statistically estimated event probabilities.

Strategy	Equity crash	Interest-rate shock	Inflation shock
Equal Weight	−13.37%	−10.38%	−7.50%
Minimum Variance	+0.38%	−1.21%	−0.30%
Maximum Sharpe	+0.09%	−1.82%	+0.35%
Risk Parity	−2.96%	−5.32%	−3.43%

Equal Weight suffered the largest modeled loss in each of the three scenarios, despite leading the historical performance comparison. This is a valuable reminder that the strategy with the best realized return need not have the best protection against a particular shock. Minimum Variance showed limited modeled losses in these scenarios. Maximum Sharpe showed small or positive instantaneous responses, despite having the worst realized historical drawdown; there is no contradiction because stress tests use a single current allocation and assumed shocks, while drawdown spans changing allocations and an extended historical path.

10. Turnover, trading costs and implementation

10.1 Why turnover matters

A strategy can generate attractive theoretical returns yet be expensive to operate. Monthly one-way turnover measures the fraction of portfolio value traded to restore target weights after market-driven drift. This makes it possible to compare how actively different optimization methods reposition capital.

Strategy	Annualized one-way turnover	Gross annualized return	Net annualized return at 10 bps
Equal Weight	11.95%	7.89%	7.87%

Minimum Variance	4.57%	1.87%	1.86%
Maximum Sharpe	326.54%	4.13%	3.48%
Risk Parity	35.82%	4.97%	4.89%

Maximum Sharpe turned over approximately 3.27 times portfolio value annually on the one-way measure, far more than the other strategies. At an illustrative 10 basis points per dollar traded, its annualized return fell by 0.65 percentage points, from 4.13% to 3.48%. The comparison reveals a meaningful implementation disadvantage that a gross-return-only presentation would conceal.

10.2 What the cost model excludes

A fixed basis-point assumption is a useful first sensitivity analysis, not a complete trading-cost model. Actual execution depends on bid–ask spreads, market impact, brokerage arrangements, ETF liquidity, order size, tax treatment and timing. Cost assumptions of 5, 10 and 25 basis points were examined in the project; detailed scenario outputs are available in the exported transaction-cost table.

11. Integrated findings and investment implications

11.1 Complexity did not guarantee superior realized performance

The strongest result was the relative success of Equal Weight. This does not invalidate optimization; it shows that a well-defined benchmark is essential and that model complexity must justify its added estimation and operational burden.

11.2 Risk cannot be reduced to one metric

Minimum Variance minimized volatility and drawdown in the observed sample, while Equal Weight delivered the strongest realized return and Sharpe. The stress scenarios and simulated loss probabilities add different dimensions of vulnerability. Portfolio selection therefore depends on the investor's objective function, risk tolerance, liability profile and implementation constraints.

11.3 Robustness and turnover deserve explicit monitoring

The Maximum Sharpe strategy's high turnover and weak relative out-of-sample performance make weight stability and estimation sensitivity priority areas for further testing. Practical controls could include turnover penalties, allocation caps, shrinkage estimates, longer estimation windows and transaction-cost-aware optimization.

11.4 Implications for an investment committee

- Use Equal Weight as a demanding, transparent benchmark rather than treating optimization as automatically superior.
- Assess Minimum Variance against explicit capital-preservation objectives and acceptable return shortfalls.

- Treat Risk Parity as a candidate for diversified risk budgeting, subject to changing correlations and bond sensitivity.
- Require Maximum Sharpe to demonstrate stability and net-of-cost improvement before deployment.
- Report realized performance, tail risk, scenario loss and turnover together rather than selecting portfolios on one headline statistic.

12. Methodological limitations

- Selection and sample dependence: eight chosen ETFs and the 2016–2025 market regime cannot establish universal strategy rankings.
- Estimation risk: 252-day historical means and covariances are imperfect inputs; expected returns are especially unstable.
- Long-only constraints: no leverage, short selling, derivatives hedging or dynamic cash allocation were evaluated.
- Simulation uncertainty: normal-return Monte Carlo can miss fat tails and structural breaks; alternative bootstrap methods have different assumptions.
- Stress-test subjectivity: scenario shocks are illustrative, not empirical event probabilities.
- Implementation frictions: fixed-basis-point costs omit detailed liquidity, market impact and taxes.
- Benchmarking: the analysis compares four internal strategies, not a full suite of institutional policy benchmarks or factor-adjusted alphas.
- Code-output consistency: the final notebook, output CSVs and report should be reconciled before external publication, particularly simulation methodology and annualization conventions.

13. Recommended extensions

The most valuable next phase is a decision-oriented dashboard that presents cumulative wealth, rolling risk, allocation weights, scenario losses and gross-versus-net results. It should be built from the existing exported data rather than introducing new unvalidated calculations.

For research depth, future experiments could compare alternative rolling-window lengths, quarterly versus monthly rebalancing, shrinkage covariance, robust expected-return estimation, turnover-penalized optimization, realistic transaction costs and expanding-window or nested walk-forward validation. Exporting monthly strategy weights would enable concentration analysis and explain why Maximum Sharpe trades so frequently.

14. Conclusion

This project demonstrates an end-to-end quantitative investment research workflow: constructing a diversified ETF universe, estimating portfolio weights, testing decisions

out of sample, quantifying downside risk, exploring modeled outcomes, applying hypothetical macroeconomic shocks and incorporating trading frictions. Its central lesson is not that one strategy is universally best. Rather, investment quality must be assessed through realized risk-adjusted results, resilience, robustness and implementation feasibility. Within the observed sample, Equal Weight delivered the strongest realized return and Sharpe, Minimum Variance best controlled volatility and drawdown, Risk Parity offered a middle-ground risk profile, and Maximum Sharpe exposed the potential cost of unstable optimization.

Appendix A. Key formulas and definitions

Measure	Definition / interpretation
Daily simple return	$P_t / P_{t-1} - 1$
Portfolio expected return	$w^T\mu$
Portfolio variance	$w^T\Sigma w$
Portfolio volatility	$\sqrt{(w^T\Sigma w)}$
Sharpe ratio	(Annualized return − risk-free rate) / annualized volatility
Drawdown	Wealth / running peak wealth − 1
Historical VaR (95%)	Loss threshold associated with lower 5% empirical return quantile
Historical CVaR (95%)	Average loss conditional on falling into worst 5% tail
Risk contribution	$w_i(\Sigma w)_i$ / portfolio volatility
Transaction-cost impact	Trading volume × assumed cost per dollar traded

Appendix B. Exported project evidence

- investment_performance_summary.csv — core comparative performance measures
- portfolio_risk_metrics.csv — portfolio risk measures and supporting metrics
- portfolio_daily_returns.csv — daily strategy return paths
- monte_carlo_summary.csv — modeled simulation summaries
- economic_stress_results.csv — scenario loss estimates
- portfolio_turnover_summary.csv — annualized trading activity
- transaction_cost_sensitivity.csv — sensitivity across cost assumptions
- portfolio_cumulative_performance.png and portfolio_drawdown_comparison.png — presentation figures

Appendix C. References and conceptual foundations

- Markowitz, H. (1952). Portfolio Selection. The Journal of Finance, 7(1), 77–91.
- Sharpe, W. F. (1966). Mutual Fund Performance. The Journal of Business, 39(1), 119–138.
- Jorion, P. (2007). Value at Risk: The New Benchmark for Managing Financial Risk (3rd ed.). McGraw-Hill.
- Meucci, A. (2005). Risk and Asset Allocation. Springer.
- Maillard, S., Roncalli, T., & Teïletche, J. (2010). The Properties of Equally Weighted Risk Contribution Portfolios. The Journal of Portfolio Management, 36(4), 60–70.
- Project source: historical ETF data downloaded using the yfinance interface to Yahoo Finance; figures should be independently validated against saved source files.